GNSS Reflectometry Channel Impulse Response Estimation Using an Extended Kalman Filter

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1 Introduction

The objectives of this report are:

- To rectify the notation used in [8], [12] and in Thiago Lobo's mathematical development.
- To propose a modified GNSS signal channel estimator based on an extended KF that tracks the state error and a control technique that provides carrier-phase estimates based on the state error estimates.

The source code of this document shall be directly used for the production of Thiago Lobo's Master's dissertation and articles that he may write.

2 Received Signal Model

A single-frequency transmitted Global Positioning System (GPS) pilot signal for a single satellite can be described by

$$s_{\mathrm{RF}}(t) = \sqrt{2P_s}c(t)\cos(2\pi f_{RF} t) \in \mathbb{R} \tag{1}$$

where P_s indicates the transmitted signal power, f_{RF} denotes the transmitted signal radiofrequency, and

$$c(t) = c(t - gT) = \sum_{r=0}^{N_c-1} a_r p(t - rT_c) \;\forall\; g \in \mathbb{Z} \tag{2}$$

symbolizes the pseudorandom (PR) code-division multiple access (CDMA) spectral spreading sequence with period T, also known as the coarse acquisition (C/A) code [11, Eq. 2.1]. Additionally, T_c denotes the code-chip period, $N_c = T/T_c$ represents the amount of chips inside a code period, and $a_r \in \{-1, 1\}$ are binary phase shift-keying chip values. The chip pulse shape is represented as [3, Eq. 4.13]

$$p(t) = \begin{cases} \dfrac{1}{\sqrt{T_c}}, & 0 \leq t \leq T_c \\ 0, & \text{elsewhere} \end{cases}, \tag{3}$$

which is assumed to be rectangular, whose continuous-time Fourier transform is given by

$$P(f) = \sqrt{T_c}\mathrm{sinc}(fT_c), \tag{4}$$

where

$$\operatorname{sinc}(\mathrm{x}) = \frac{\sin(\pi \mathrm{x})}{\pi \mathrm{x}}. \tag{5}$$

The autocorrelation (ACF), its derivative, and the power spectral density (PSD) of $p(t)$ is therefore given as [3, Eq. 4.14] I think that the autocorrelation function of the pulse shape only corresponds to the function shown in Kaplan in the case when we evaluate it within $\left[-\frac{T_c}{2}, \frac{T_c}{2}\right]$. I think that there is a phase shift factor missing in the textbook definition. However, it is fine to reference to the textbook equation as it is.

$$\begin{aligned}\Phi_{pp}(\tau) &= \int_T p(t)\, p(t+\tau)\, \mathrm{d}t \\ &= \begin{cases} 1 - \frac{|\tau|}{T_c}, & |\tau| \leq T_c \\ 0, & \text{elsewhere} \end{cases},\end{aligned} \tag{6}$$

and

$$S_{pp}(f) = T_c \operatorname{sinc}^2(f T_c), \tag{7}$$

respectively.

It is noteworthy that although other kinds of chip pulse shape are also used in modern GNSS signals, as discussed in [4, Section 4.2.3], the rectangular pulse shape is widely adopted.

Considering the Global Navigation Satellite System Reflectometry (GNSS-R), the transmitted signal travels from a GNSS satellite through the ionosphere and the troposphere until it reaches the ground and then it is reflected. After reflection, the signal passes again through both the troposphere and some layers of the ionosphere until it reaches an on-board antenna of a low-earth orbit (LEO) satellite.

In the sequel, we approximate the channel effects, by the tapped-delay line (TDL) [12, Eq. 3]

$$h(t, \tau) = \sum_{L}^{l=0} h_l(t)\, \delta(\tau - \tau_{\mathrm{T}}(t) - l T_s) \in \mathbb{C} \tag{8}$$

where $L = 2q + 1$ denotes the amount of channel taps, with $q \in \mathbb{N}$, $\alpha_l(t) \in \mathbb{C}$ represent the complex time-varying channel coefficients, $\delta(\cdot)$ is the Dirac delta function, and $\delta\tau$ is the delay-grid spacing. The *total apparent code-delay* and is modeled as

$$\tau_{\mathrm{T}}(t) = \tau_{\mathrm{LOS}}(t) + \tau_{\mathrm{iono}, f_{\mathrm{RF}}}(t) + \tau_{\mathrm{tropo}}(t) + \tau_{\mathrm{other}}(t) \in \mathbb{R}. \tag{9}$$

with geometric bistatic component

$$\tau_{\mathrm{LOS}}(t) = \frac{r_{\mathrm{T_x} \rightarrow \mathrm{patch}}(t) + r_{\mathrm{patch} \rightarrow \mathrm{R_x}}(t)}{c_0}, \tag{10}$$

where $r_{\mathrm{T_x} \rightarrow \mathrm{patch}}(t)$ and $r_{\mathrm{patch} \rightarrow \mathrm{R_x}}(t)$ represent the transmitter to ground-patch and the ground-patch to receiver time-varying distances, respectively, and c_0 models the speed of light at vacuum. The two-way time-delays of the dispersive ionosphere and the non-dispersive troposphere are represented as $\tau_{\mathrm{iono}, f_{\mathrm{RF}}}(t)$ and $\tau_{\mathrm{tropo}}(t)$, respectively, while the other unmodeled sources of time-delay are modeled by $\tau_{\mathrm{other}}(t)$.

In this view, we model the continuous-time GNSS received signal as the convolution, represented by the symbol $*$, of the transmitted signal (1) with the TDL represented by (8), with thermal noise $v(t)$ (assumed to be temporally white, Gaussian, and weak-sense stationary) with power spectral density N_0, yielding

$$\begin{aligned} y_{\mathrm{RF}}(t) &= (s_{\mathrm{RF}} * h)(t) + v(t) && (11)\\ &= \int_{-\infty}^{+\infty} s_{\mathrm{RF}}(t-\tau)\, h(t,\tau)\, \mathrm{d}\tau + v(t) && (12)\\ &= \int_{-\infty}^{+\infty} s_{\mathrm{RF}}(t-\tau) \sum_{l=0}^{L} h_l(t)\, \delta(\tau - \tau_{\mathrm{T}}(t) - lT_s)\, \mathrm{d}\tau + v(t). && (13) \end{aligned}$$

Rearranging the terms and noticing that the terms of the integral only yields non-zero values for $\tau = \tau_{\mathrm{T}}(t) + lT_s$, we get

$$y_{\mathrm{RF}}(t) = \sum_{l=0}^{L} h_l(t)\, s_{\mathrm{RF}}(t - \tau_{\mathrm{T}}(t) - lT_s) + v(t) \in \mathbb{C}. \tag{14}$$

Using the fundamental theorem of calculus [5, Section 5.3], the total phase-equivalent delay evolves as

$$\tau_{\mathrm{T}}(t) = \tau_{\mathrm{T}}(t_0) + \int_{t_0}^{t} \frac{\partial \tau_{\mathrm{T}}(u)}{\partial u}\, du \tag{15}$$

Substituting (15) into (14) gives

$$y_{\mathrm{RF}}(t) = \sum_{l=0}^{L} h_l(t) \sqrt{2P_s}\, c(t - \tau_{\mathrm{T}}(t) - lT_s) \cos(2\pi f_{\mathrm{RF}} t + \phi_{\mathrm{T}}(t) + \phi_l) + v(t), \tag{16}$$

where

$$\phi_{\mathrm{T}}(t_0) = -2\pi f_{\mathrm{RF}} \tau_{\mathrm{T}}(t_0) \tag{17}$$

represents the initial total phase shift,

$$f_T(t) = -f_{\mathrm{RF}} \frac{\partial \tau_{\mathrm{T}}(t)}{\partial t} \tag{18}$$

indicates the total frequency shift, and

$$\phi_{\mathrm{T}}(t) = \phi_{\mathrm{T}}(t_0) + 2\pi \int_{t_0}^{t} f_T(u)\, du \tag{19}$$

denotes the total phase evolution. The per-tap constant phase is

$$\phi_l = -2\pi f_{\mathrm{RF}} l T_s. \tag{20}$$

Therefore, we can clearly observe that the time-delay introduced by the channel effects yields a time-varying phase shift in the carrier.

The continuous-time RF signal is band-select filtered and amplified by a low-noise amplifier (LNA). It is then mixed with a local oscillator (LO) to produce either a low-IF or zero-IF signal. A channel-select filter removes the mixer sum term and out-of-band products, and the signal is sampled by the analog-to-digital converter (ADC) at a $f_s = \frac{1}{T_s} = 2B$ rate that satisfies the Nyquist criterion for the selected bandwidth. In our setup we intentionally do not use an automatic gain control (AGC) to preserve the amplitude features of the received signal.

The sampled IF is multiplied by a complex sinusoid provided by a numerically controlled oscillator (NCO) to shift it to baseband. The I/Q streams are low-pass filtered to reject the high-frequency images. The signal is then decimated to the processing rate. At this point, we assume that both the analog preprocessing and digital downconversion steps did not introduce any artifacts to the received signal. Therefore, we may model the discrete-time received signal as

$$y\left[kN+n\right]=\sum_{l=0}^{L}h_{l}\left[k\right]\sqrt{2P_{s}}c\left(\left(kN+n\right)T_{s}-\tau_{\mathrm{T}}\left[k\right]-lT_{s}\right)e^{j\left(\phi_{\mathrm{T}}\left[kN+n\right]+\phi_{l}\right)}+v\left[kN+n\right], \tag{21}$$

where $k\in\{0,1,\ldots,K-1\}$ represents the index of PR-code block in the signal stream, $n\in\{0,1,\ldots,N-1\}$ indicates the sample index within a PR-code block, where $N=T/T_s$. Here, we assumed that the total apparent code-delay $\tau_{\mathrm{T}}\left(t\right)$, and the channel coefficients $h_l\left(t\right)$ are approximately constant within a k-th interval, so that we can state

$$h_{l}\left[k\right]=h_{l}\left(\left(kN+N-1\right)T_{s}\right)\approx h_{l}\left(\left(kN+n\right)T_{s}\right)\ \forall\ n, \tag{22}$$
$$\tau_{\mathrm{T}}\left[k\right]=\tau_{\mathrm{T}}\left(\left(kN+N-1\right)T_{s}\right)\approx\tau_{\mathrm{T}}\left(\left(kN+n\right)T_{s}\right)\ \forall\ n. \tag{23}$$

Moreover, we have that

$$\phi_{\mathrm{T}}\left[kN+n\right]=\phi_{\mathrm{T}}\left(\left(kN+n\right)T_{s}\right) \tag{24}$$

and

$$v\left[kN+n\right]=v\left(\left(kN+n\right)T_{s}\right) \tag{25}$$

denote the discretized version of $\phi_{\mathrm{T}}\left(t\right)$ and $v\left(t\right)$, respectively. The variance of the discrete-time thermal noise is assumed to be time-invariant and given by $\sigma_v^2=2BN_0$.

For a compact notation, we introduce the phase-shifted channel coefficients

$$\tilde{h}_{l}\left[k\right]=h_{l}\left[k\right]e^{j\phi_{l}}. \tag{26}$$

which can be represented as a convolution matrix with a circulant Toeplitz structure

$$\mathbf{H}\left[k\right]=\sum_{l=0}^{L}\tilde{h}_{l}\left[k\right]\mathbf{P}^{l}, \tag{27}$$

where $\mathbf{P}\in\mathbb{R}^{N\times N}$ is the cyclic right-shift matrix

$$\mathbf{P}=\begin{bmatrix}\mathbf{0}_{N-1}^{\top} & 1\\ \mathbf{I}_{N-1} & \mathbf{0}_{N-1}\end{bmatrix}. \tag{28}$$

We now introduce the following vectorized forms of the discrete-time signals

$$\mathbf{d}\left[k\right]=\left[e^{j\phi_{\mathrm{T}}\left[kN\right]},e^{j\phi_{\mathrm{T}}\left[kN+1\right]},\ldots,e^{j\phi_{\mathrm{T}}\left[kN+N-1\right]}\right]^{\top}, \tag{29}$$
$$\mathbf{c}\left(\tau_{\mathrm{T}}\left[k\right]\right)=\left[c\left(\left(kN\right)T_{s}-\tau_{\mathrm{T}}\left[k\right]\right),c\left(\left(kN+1\right)T_{s}-\tau_{\mathrm{T}}\left[k\right]\right),\right.$$
$$\left.\ldots,c\left(\left(kN+N-1\right)T_{s}-\tau_{\mathrm{T}}\left[k\right]\right)\right]^{\top}, \tag{30}$$
$$\mathbf{v}\left[k\right]=\left[v\left[kN\right],v\left[kN+1\right],\ldots,v\left[kN+N-1\right]\right]^{\top}. \tag{31}$$

The signal (21) can be then reformulated as

$$\begin{aligned}\mathbf{y}[k] &= [y[kN], y[kN+1], \ldots, y[kN+N-1]]^\top \\ &= \sqrt{2P_s}\left(\mathbf{H}[k]\,\mathbf{c}(\tau_\mathrm{T}[k])\right) \odot \mathbf{d}[k] + \mathbf{v}[k],\end{aligned} \tag{32}$$

where

$$\mathbf{v}[k] \sim \mathcal{N}\left(\mathbf{0}_N, \sigma_v^2 \mathbf{I}_N\right) \tag{33}$$

is modeled as a temporally uncorrelated, multivariate Gaussian process.

It is noteworthy the free-space path loss due to the time-varying distance between the receiver and the satellite is fully captured by the channel coefficients. (Rodrigo): We can later add a free-space loss (e.g., $\gamma(t)$) scaling factor to equation (41), so the channel coefficients are properly scaled with respect with the received signal power.

3 Signal Processing

In this Section, we will discuss the details of the proposed digital signal processor, which enables the channel impulse response estimation of GNSS-R applications. Here, we divided it in three separate Subsections: the Carrier Dynamics Removal, the Multicorrelator Bank, the EKF Algorithm, and the NCO and code replica generator, which comprises each step of the proposed system. (Rodrigo): Thiago, note that in this report, i decided to separate the carrier-phase dynamics removal from the multicorrelator bank, for improving the clarity for the reader. Since there is now the numerically controlled oscillator, i understood that it would be clearer to disjoint the carrier-phase correction step from the multicorrelator bank.

The first step of signal processing consists in correcting the total appearent phase shift in the received signal by performing an element-wise multiplication with a carrier replica with conjugated phase dynamics. This carrier replica is provided by an NCO, which uses carrier-phase, Doppler shift and Doppler drift dynamics estimates from an EKF, which are going to be discussed in details in Subsection 3.2.

Therefore, we may model the carrier replica samples in a vectorized form as

$$\hat{\mathbf{d}}[k] = \left[e^{j\hat{\phi}_\mathrm{T}[kN]}, e^{j\hat{\phi}_\mathrm{T}[kN+1]}, \ldots, e^{j\hat{\phi}_\mathrm{T}[kN+N-1]}\right]^\top \tag{34}$$

After element-wise multiplying the Hermitian (i.e., the conjugated transpose) of 34 with 32, we get

$$\begin{aligned}\tilde{\mathbf{y}}[k] &= \left(\sqrt{2P_s}\left(\mathbf{H}[k]\,\mathbf{c}(\tau_\mathrm{T}[k])\right) \odot \mathbf{d}[k] + \mathbf{v}[k]\right) \odot \hat{\mathbf{d}}^\mathsf{H}[k] \\ &= \sqrt{2P_s}\left(\mathbf{H}[k]\,\mathbf{c}(\tau_\mathrm{T}[k])\right) \odot \left(\mathbf{d}[k] \odot \hat{\mathbf{d}}^\mathsf{H}[k]\right) + \mathbf{v}[k] \odot \hat{\mathbf{d}}^\mathsf{H}[k] \qquad (35) \\ &= \sqrt{2P_s}\left(\mathbf{H}[k]\,\mathbf{c}(\tau_\mathrm{T}[k])\right) \odot \tilde{\mathbf{d}}[k] + \mathbf{v}[k] \qquad (36)\end{aligned}$$

where $(\cdot)^\mathsf{H}$ denotes the Hermitian operator and

$$\begin{aligned}\tilde{\mathbf{d}}[k] &= \left[e^{j\left(\phi_\mathrm{T}[kN]-\hat{\phi}_\mathrm{T}[kN]\right)}, e^{j\left(\phi_\mathrm{T}[kN+1]-\hat{\phi}_\mathrm{T}[kN+1]\right)}, \ldots, e^{j\left(\phi_\mathrm{T}[kN+N-1]-\hat{\phi}_\mathrm{T}[kN+N-1]\right)}\right]^\top \\ &= \left[e^{j\varepsilon_\phi[kN]}, e^{j\varepsilon_\phi[kN+1]}, \ldots, e^{j\varepsilon_\phi[kN+N-1]}\right]^\top\end{aligned} \tag{37}$$

represents the residual carrier samples, with

$$\varepsilon_\phi[kN+n] = \phi_\mathrm{T}[kN+n] - \hat{\phi}_\mathrm{T}[kN+n]. \tag{38}$$

In the passage from 35 to 36, we used the fact that phase shifts in $\mathbf{v}[k]$ do not affect its second-order statistics, given the assumption of weak-sense stationarity [2].

In the sequel, we perform multiple correlations with shifted replicas of the PR-code sequence, as proposed in [8, Section 3] and in [12, Section 2]. The multicorrelator bank with $C = 2q+1$ code replicas, with $q \in \mathbb{N}$, can be represented as

$$\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right) = \begin{bmatrix} \mathbf{c}^\top\left(\hat{\tau}_\mathrm{T}[k] + q\Delta_\tau\right) \\ \vdots \\ \mathbf{c}^\top\left(\hat{\tau}_\mathrm{T}[k]\right) \\ \vdots \\ \mathbf{c}^\top\left(\hat{\tau}_\mathrm{T}[k] - q\Delta_\tau\right) \end{bmatrix} \in \mathbb{R}^{C\times N} \tag{39}$$

where $\hat{\tau}_\mathrm{T}[k]$ represents the code-delay provided by the LQG controller, which is going to be characterized later in this work.

A practical EKF initialization can rely on (i) the time-delay estimate from acquisition, or (ii) a coarse geometric range computed from known positions: the GNSS satellite from broadcast ephemeris; the LEO satellite from its precise-orbit-determination (POD) solution; and the specular reflection point on Earth from a mean sea-surface model [14, Section II-A], such as DTU18 [10]. From these, compute the free-space propagation time from the GNSS transmitter to the LEO receiver and use it as the initial time delay.

After multiplying (39) in the left-side of (36), it yields

$$\begin{aligned} \mathbf{z}[k] &= \frac{1}{N}\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right)\tilde{\mathbf{y}}[k] \\ &= \frac{1}{N}\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right)\left(\sqrt{2P_s}\left(\mathbf{H}[k]\,\mathbf{c}\left(\tau_\mathrm{T}[k]\right)\right)\odot\tilde{\mathbf{d}}[k] + \mathbf{v}[k]\right) \\ &= \frac{\sqrt{2P_s}}{N}\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right)\left(\left(\mathbf{H}[k]\,\mathbf{c}\left(\tau_\mathrm{T}[k]\right)\right)\odot\tilde{\mathbf{d}}[k]\right) + \frac{1}{N}\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right)\mathbf{v}[k]\,. \end{aligned} \tag{40}$$

Noticing that

$$\mathbf{H}[k]\,\mathbf{c}\left(\tau_\mathrm{T}[k]\right)\odot\tilde{\mathbf{d}}[k] = \sum_{l=0}^{L}\tilde{h}_l[k]\,\mathbf{c}\left(\tau_\mathrm{T}[k] + lT_s\right)\odot\tilde{\mathbf{d}}[k]\,, \tag{41}$$

adopting

$$\mathbf{v}_C[k] = \frac{1}{N}\mathbf{C}^\top\left(\hat{\tau}_\mathrm{T}[k]\right)\mathbf{v}[k] \in \mathbb{C}^C \tag{42}$$

for the colored noise, and substituting (41) and (42) into (40), we get

$$\begin{aligned}
\mathbf{z}[k] &= \frac{\sqrt{2P_s}}{N}\begin{bmatrix}\mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]+q\Delta_\tau)\\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]) \\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]-q\Delta_\tau)\end{bmatrix}\left(\sum_{l=0}^{L}\tilde{h}_l[k]\,\mathbf{c}(\tau_\mathrm{T}[k]+lT_s)\odot\tilde{\mathbf{d}}[k]\right)+\mathbf{v}_C[k]\\
&= \frac{\sqrt{2P_s}}{N}\sum_{l=0}^{L}\tilde{h}_l[k]\begin{bmatrix}\mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]+q\Delta_\tau)\left(\mathbf{c}(\tau_\mathrm{T}[k]+lT_s)\odot\tilde{\mathbf{d}}[k]\right)\\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k])\left(\mathbf{c}(\tau_\mathrm{T}[k]+lT_s)\odot\tilde{\mathbf{d}}[k]\right) \\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]-q\Delta_\tau)\left(\mathbf{c}(\tau_\mathrm{T}[k]+lT_s)\odot\tilde{\mathbf{d}}[k]\right)\end{bmatrix}+\mathbf{v}_C[k]\\
&= \frac{\sqrt{2P_s}}{N}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\begin{bmatrix}\chi(\tau_1(l),\tau_2(-q),\varepsilon_\nu[kN])\\ \vdots \\ \chi(\tau_1(l),\tau_2(0),\varepsilon_\nu[kN]) \\ \vdots \\ \chi(\tau_1(l),\tau_2(q),\varepsilon_\nu[kN])\end{bmatrix}+\mathbf{v}_C[k], \qquad (43)
\end{aligned}$$

where $\tau_1(l)=\tau_\mathrm{T}[k]+lT_s$ and $\tau_2(m)=\hat{\tau}_\mathrm{T}[k]-m\Delta_\tau$ and the ambiguity function is defined as

$$\begin{aligned}
&\chi(\tau_1(l),\tau_2(m),\varepsilon_\nu[kN]) =\\
&= e^{-j\varepsilon_\phi[kN]}\mathbf{c}^\top(\tau_2(m))\left(\mathbf{c}(\tau_1(l))\odot\tilde{\mathbf{d}}[k]\right)\\
&= e^{-j\varepsilon_\phi[kN]}\sum_{n=0}^{N-1}c((kN+n)T_s-\tau_1(l))\,c((kN+n)T_s-\tau_2(m))\,e^{j\varepsilon_\phi[kN+n]} \qquad (44)\\
&= \sum_{n=0}^{N-1}c((kN+n)T_s-\tau_1(l))\,c((kN+n)T_s-\tau_2(m))\,e^{j2\pi\varepsilon_\nu[kN]nT_s} \qquad (45)\\
&\forall\, m\in\{-q,\ldots,0,\ldots,q\}.
\end{aligned}$$

In the passage from (44) to (45), we modeled the instantaneous carrier-phase error $\varepsilon_\phi[kN+n]$ as the sum of the initial phase error inside the PR-code block with a phase ramp due to a residual frequency shift $\varepsilon_\nu[kN]$, i.e.

$$\varepsilon_\phi[kN+n]=\varepsilon_\phi[kN]+2\pi\varepsilon_\nu[kN]\,nT_s. \qquad (46)$$

Invoking the assumption that $T_s \ll T_c$, we may describe the frequency shift weighted discrete correlation (45) analytically as a continuous-time correlation, such that

$$\chi(\tau_1(l),\tau_2(m),\varepsilon_\nu[kN]) \approx \int_T c(t-\tau_1(l))\,c(t-\tau_2(m))\,e^{j2\pi\varepsilon_\nu[kN]t}\mathrm{d}t. \qquad (47)$$

Substituing (2) into (47), and considering that the correlation is performed for N samples, as shown in

(45), we get

$$
\begin{aligned}
&\chi\left(\tau_1(l), \tau_2(m), \varepsilon_\nu[kN]\right) \approx \\
&\approx \int_T \left(\sum_{r=0}^{N-1} a_r p\left(t-\tau_1(l)-rT_s\right)\right)\left(\sum_{s=0}^{N-1} a_s p\left(t-\tau_2(m)-sT_s\right)\right) e^{j2\pi\varepsilon_\nu[kN]t}\mathrm{d}t \\
&\approx \sum_{r=0}^{N-1}\sum_{s=0}^{N-1} a_r a_s \int_T p\left(t-\tau_1(l)-rT_s\right) p\left(t-\tau_2(m)-sT_s\right) e^{j2\pi\varepsilon_\nu[kN]t}\mathrm{d}t. \qquad (48)
\end{aligned}
$$

Substituing $u = t-\tau_1(l)-rT_s$, noticing that $\mathrm{d}u = \mathrm{d}t$, and assuming that $a_r a_s = 1$ for $r = s$, it yields

$$
\begin{aligned}
&\chi\left(\tau_1(l), \tau_2(m), \varepsilon_\nu[kN]\right) \approx \\
&\approx \sum_{r=0}^{N-1}\sum_{s=0}^{N-1} a_r a_s \int_T p(u)\, p\left(u+\tau_1(l)+rT_s-\tau_2(m)-sT_s\right) e^{j2\pi\varepsilon_\nu[kN](u+\tau_1(l)+rT_s)}\mathrm{d}u \\
&\approx \sum_{r=0}^{N-1} \int_T p\left(u+\tau_1(l)-\tau_2(m)\right) p(u)\, e^{j2\pi\varepsilon_\nu[kN](u+\tau_1(l)+rT_s)}\mathrm{d}u \\
&\approx \underbrace{e^{j2\pi\varepsilon_\nu[kN]\tau_1(l)}}_{\approx 1} \underbrace{\sum_{r=0}^{N-1} e^{j2\pi\varepsilon_\nu[kN]rT_s}}_{\approx N} \underbrace{\int_T p(u)\, p\left(u+\tau_1(l)-\tau_2(m)\right) e^{j2\pi\varepsilon_\nu[kN]u}\mathrm{d}u}_{\approx \Phi_{pp}(\tau_1(l)-\tau_2(m)),(6)} \qquad (49)
\end{aligned}
$$

For $\varepsilon_\nu[kN] \ll 1$, we can further approximate the ambiguity function as the autocorrelation function of $c(t)$ evaluated in $\tau_1(l)-\tau_2(m)$, i.e., $\Phi_{cc}\left(\tau_1(l)-\tau_2(m)\right)$ such that

$$
\begin{aligned}
\chi\left(\tau_1(l), \tau_2(m), 0\right) &\approx \Phi_{cc}\left(\tau_1(l)-\tau_2(m)\right) \qquad (50) \\
&\approx N\Phi_{pp}\left(\tau_1(l)-\tau_2(m)\right) \\
&\approx \begin{cases} N\left(1-\dfrac{|\tau_1(l)-\tau_2(m)|}{T_c}\right), & |\tau_1(l)-\tau_2(m)| \le T_c \\ 0, & \text{otherwise} \end{cases}. \qquad (51)
\end{aligned}
$$

(Rodrigo): Thiago, if the frequency shift error is high, however it would be necessary to deal with (49) analytically, as you have done in your mathematical development. However, it is also reasonable to assume that the residual frequency shift is very low, since this is commonly the case in GNSS receivers. If we observe that frequency shift error is high in our simulations, cranking up the values of the covariance matrix of the carrier-phase dynamics should help.

Substituing (50) into (43), it yields

$$
\mathbf{z}[k] = \sqrt{2P_s}e^{j\varepsilon_\phi[kN]} \sum_{l=0}^{L} \tilde{h}_l[k] \begin{bmatrix} \Phi_{pp}\left(\tau_1(l)-\tau_2(-q)\right) \\ \vdots \\ \Phi_{pp}\left(\tau_1(l)-\tau_2(0)\right) \\ \vdots \\ \Phi_{pp}\left(\tau_1(l)-\tau_2(q)\right) \end{bmatrix} + \mathbf{v}_C[k], \qquad (52)
$$

which is a more generic version of [12, Eq. 18], since we have a explicit power scaling and a residual phase error shift factors, as well as a timing residual, i.e., we do not assume that $\tau_\mathrm{T}[k]-\hat{\tau}_\mathrm{T}[k] \approx 0$.

In the special case where $\Delta_\tau = T_s$, which is adopted in [12], we may resubstitute $\tau_1(l)$ and $\tau_2(m)$ and we would get

$$\begin{aligned}
\mathbf{z}[k] &= \sqrt{2P_s}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\begin{bmatrix}\Phi_{pp}\left(\tau_{\mathrm{T}}[k]+lT_s-\left(\hat{\tau}_{\mathrm{T}}[k]+qT_s\right)\right)\\ \vdots \\ \Phi_{pp}\left(\tau_{\mathrm{T}}[k]+lT_s-\hat{\tau}_{\mathrm{T}}[k]\right)\\ \vdots \\ \Phi_{pp}\left(\tau_{\mathrm{T}}[k]+lT_s-\left(\hat{\tau}_{\mathrm{T}}[k]-qT_s\right)\right)\end{bmatrix}+\mathbf{v}_C[k],\\
&= \sqrt{2P_s}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\begin{bmatrix}\Phi_{pp}\left(\varepsilon_\tau[k]+(l-q)T_s\right)\\ \vdots \\ \Phi_{pp}\left(\varepsilon_\tau[k]+lT_s\right)\\ \vdots \\ \Phi_{pp}\left(\varepsilon_\tau[k]+(l+q)T_s\right)\end{bmatrix}+\mathbf{v}_C[k]\\
&= \sqrt{2P_s}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\,\mathbf{\Phi}_{pp}\left(\varepsilon_\tau[k],l\right)+\mathbf{v}_C[k]\in\mathbb{C}^C
\end{aligned}\tag{53}$$

where $\varepsilon_\tau[k] = \tau_{\mathrm{T}}[k] - \hat{\tau}_{\mathrm{T}}[k]$, and

$$\mathbf{\Phi}_{pp}\left(\varepsilon_\tau[k],l\right) = \begin{bmatrix}\Phi_{pp}\left(\varepsilon_\tau[k]+(l-q)T_s\right)\\ \vdots \\ \Phi_{pp}\left(\varepsilon_\tau[k]+lT_s\right)\\ \vdots \\ \Phi_{pp}\left(\varepsilon_\tau[k]+(l+q)T_s\right)\end{bmatrix}\in\mathbb{R}^C. \tag{54}$$

Furthermore, we may derive the covariance matrix of the colored noise $\mathbf{v}_C[k]$ as

$$\mathbf{R} = \mathbb{E}\left\{\left(\mathbf{v}_C[k]-\mathbb{E}\left\{\mathbf{v}_C[k]\right\}\right)\left(\mathbf{v}_C[k]-\mathbb{E}\left\{\mathbf{v}_C[k]\right\}\right)^{\mathsf{H}}\right\} = \mathbb{E}\left\{\mathbf{v}_C[k]\,\mathbf{v}_C^{\mathsf{H}}[k]\right\} \tag{55}$$

where $\mathbb{E}\{\cdot\}$ denotes the expected value operator. Here, we invoked the zero-mean property of $\mathbf{v}_C[k]$, i.e., $\mathbb{E}\{\mathbf{v}_C[k]\} = \mathbf{0}_C$. This is reasonable, if we assume that $v[kN+n]$ and $c((kN+n)T_s - \hat{\tau}_{\mathrm{T}}[k] - mT_s)$ are independent, and that $\mathbb{E}\{c((kN+n)T_s - \hat{\tau}_{\mathrm{T}}[k] - mT_s)\} \approx 0$. Moreover, substituing (42) into (55), using the fact that $\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])$ is real, and introducing the covariance of (33), we get

$$\begin{aligned}
\mathbf{R} &= \mathbb{E}\left\{\left(\frac{1}{N}\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\mathbf{v}[k]\right)\left(\frac{1}{N}\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\mathbf{v}[k]\right)^\mathsf{H}\right\} \\
&= \frac{1}{N^2}\mathbb{E}\left\{\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\mathbf{v}[k]\,\mathbf{v}^\mathsf{H}[k]\,\mathbf{C}(\hat{\tau}_\mathrm{T}[k])\right\} \\
&= \frac{1}{N^2}\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\mathbb{E}\left\{\mathbf{v}[k]\,\mathbf{v}^\mathsf{H}[k]\right\}\mathbf{C}(\hat{\tau}_\mathrm{T}[k]) \\
&= \frac{1}{N^2}\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\sigma_v^2\mathbf{I}_N\mathbf{C}(\hat{\tau}_\mathrm{T}[k]) \\
&= \frac{\sigma_v^2}{N^2}\mathbf{C}^\top(\hat{\tau}_\mathrm{T}[k])\,\mathbf{C}(\hat{\tau}_\mathrm{T}[k]) \\
&= \frac{\sigma_v^2}{N^2}\begin{bmatrix}\mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]+q\Delta_\tau)\\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]) \\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]-q\Delta_\tau)\end{bmatrix}\begin{bmatrix}\mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]+q\Delta_\tau)\\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]) \\ \vdots \\ \mathbf{c}^\top(\hat{\tau}_\mathrm{T}[k]-q\Delta_\tau)\end{bmatrix}^\top \\
&= \frac{\sigma_v^2}{N}\begin{bmatrix}\Phi_{pp}(0) & \cdots & \Phi_{pp}(2q\Delta_\tau)\\ \vdots & \cdots & \vdots \\ \Phi_{pp}(2q\Delta_\tau) & \cdots & \Phi_{pp}(0)\end{bmatrix}
\end{aligned} \tag{56}$$

The post-correlation signal model $\mathbf{z}[k]$ is, therefore, fully characterized and can now be used by the EKF algorithm, discussed in Subsection 3.2, for estimating the time-delay, the carrier-phase dynamics and the channel coefficients.

3.1 Proposed State-Space Model

This section explain in details the adopted models for the delay and carrier-phase dynamics, as well as the channel coefficients. The proposed models are then used in the EKF algorithm discussed in Subsection 3.2, in order to obtain meaningful state estimates.

3.1.1 Code-Delay and Carrier-Phase dynamics model

We propose to represent the code-delay and the carrier-phase dynamics as a Wiener model, which can be understood as a integrated random walk that is driven by a multivariate Gaussian process. For further details about this model, the reader is referred to [11, Section 2.4] and [17, Section 4.3.1][1]. In this work, we adopt the same notation as used by Brown and Hwang [6] Thus, we have

$$\mathbf{x}_\mathrm{W}[k] = \mathbf{F}_\mathrm{W}\mathbf{x}_\mathrm{W}[k-1] + \mathbf{w}_\mathrm{W}[k], \tag{57}$$

where

$$\mathbf{x}_\mathrm{W}[k] = [\tau_\mathrm{T}[k], \phi_\mathrm{T}[k], \nu_\mathrm{T}[k], \dot{\nu}_\mathrm{T}[k]]^\top \in \mathbb{R}^4, \tag{58}$$

[1](Rodrigo): You may also see the chapter Section 4.1 of my Master's dissertation. I did not put it as reference here, because it has note been yet published.

is the state variable, where each entry is represented in seconds, rad, rad/s and rad/s^2, respectively. Moreover,

$$\mathbf{F}_{\mathrm{W}} = \begin{bmatrix} 1 & 0 & \beta T & \beta T^2 \\ 0 & 1 & T & T^2 \\ 0 & 0 & 1 & T \\ 0 & 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{4\times 4} \tag{59}$$

denotes the Wiener model state transition matrix, where $\beta = \dfrac{1}{2\pi f_{\mathrm{RF}}}$ and

$$\mathbf{w}_{\mathrm{W}}[k] \sim \mathcal{N}(\mathbf{0}_4, \mathbf{Q}_{\mathrm{W}}) \tag{60}$$

represents the driving multivariate Gaussian random variable, whose covariance matrix can be computed using [17, Eq. 38], and is given by[2]

$$\begin{aligned}\mathbf{Q}_{\mathrm{W}} = \sigma_1^2 \begin{bmatrix} T & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \sigma_2^2 \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & T & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \sigma_3^2 \begin{bmatrix} \frac{T^3\beta^2}{3} & \frac{T^3\beta}{3} & \frac{T^2\beta}{2} & 0 \\ \frac{T^3\beta}{3} & \frac{T^3}{3} & \frac{T^2}{2} & 0 \\ \frac{T^2\beta}{2} & \frac{T^2}{2} & T & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ + \sigma_4^2 \begin{bmatrix} \frac{T^5\beta^2}{20} & \frac{T^5\beta}{20} & \frac{T^4\beta}{8} & \frac{T^3\beta}{6} \\ \frac{T^5\beta}{20} & \frac{T^5}{20} & \frac{T^4}{8} & \frac{T^3}{6} \\ \frac{T^4\beta}{8} & \frac{T^4}{8} & \frac{T^3}{3} & \frac{T^2}{2} \\ \frac{T^3\beta}{6} & \frac{T^3}{6} & \frac{T^2}{2} & T \end{bmatrix},\end{aligned} \tag{61}$$

where $\left\{\sigma_i^2\right\}_{i=1}^4 \in \mathbb{R}^+$

Note that we used here a fixed integration period T, which corresponds the period of a full PR-code block.

3.1.2 The Linear Quadratic Gaussian Controller and the Error Dynamics Modeling

Consistent with prior work, [16, Section IV-B], [15, Section 5.1.2], [11, Section III], [13, Sections 3.1 & 3.4], [7, Section 3], we formulate the code-delay and the carrier-phase dynamics EKF in an error-state form, tracking the error states rather than the absolute ones. In parallel with the Wiener model, we employ a Linear Quadratic Gaussian (LQG) controller to estimate the carrier-phase dynamics using the error-state estimate provided by the EKF as input.

Therefore, we may model the LQG controller system as [16, Eq. 52]

$$\mathbf{x}_{\mathrm{LQG}}[k] = \mathbf{F}_{\mathrm{LQG}}\mathbf{x}_{\mathrm{LQG}}[k-1] + \mathbf{B}_{\mathrm{LQG}}\mathbf{u}_{\mathrm{LQG}}[k-1] \tag{62}$$

where

$$\mathbf{x}_{\mathrm{LQG}}[k] = \left[\tau_{\mathrm{LQG}}[k], \phi_{\mathrm{LQG}}[k], \nu_{\mathrm{LQG}}[k], \dot{\nu}_{\mathrm{LQG}}[k]\right]^\top \tag{63}$$

is the controller's state variable, $\mathbf{F}_{\mathrm{LQG}} = \mathbf{F}_{\mathrm{W}}$ is the controller's state transition matrix, $\mathbf{B}_{\mathrm{LQG}} = \mathbf{I}_4$ is the control input matrix and [16, Eq. 58]

$$\mathbf{u}_{\mathrm{LQG}}[k] = \mathbf{L}\delta\mathbf{x}_{\mathrm{W}}[k] \tag{64}$$

[2](Rodrigo): Run this script that i pushed into your github repository. It shows the complete structure of the matrices discussed here, and how to obtain them.

represents the control input vector, that is composed by the control matrix $\mathbf{L}$ and the error-state vector $\delta\mathbf{x}_{\mathrm{W}}[k]$, which is assumed to be effectively provided by the EKF algorithm.

Accordingly to [16, Eq. 59], considering the separation principle [1], and subtracting (62) from (57), we have that the error-state dynamics yields:

$$\delta\mathbf{x}_{\mathrm{W}}[k] = (\mathbf{F}_{\mathrm{W}} - \mathbf{L})\,\delta\mathbf{x}_{\mathrm{W}}[k-1] + \mathbf{w}_{\mathrm{W}}[k-1], \tag{65}$$

where

$$\delta\mathbf{x}_{\mathrm{W}}[k] = \left[\varepsilon_{\tau}[k], \varepsilon_{\phi}[k], \varepsilon_{\nu}[k], \varepsilon_{\dot{\nu}}[k]\right]^{\top} \tag{66}$$

is the Wiener model error-state. Here, $\varepsilon_{\dot{\nu}}[k] = \dot{\nu}[k] - \hat{\dot{\nu}}[k]$ is the frequency drift error.

It is noteworthy that $\mathbf{L}$ should be designed in order to place the eigenvalues of $(\mathbf{F}_{\mathrm{W}} - \mathbf{L})$ at a specific group of chosen poles. This can be achieved by solving the algebraic Ricatti equation, as discussed in [13, Sections 3.2 & 3.4]. The computation of $\mathbf{L}$ is given by [13, Eq. 3.211] [3]

$$\mathbf{L} = \left(\mathbf{T}_u + \mathbf{B}_{\mathrm{LQG}}^{\top}\mathbf{S}\mathbf{B}_{\mathrm{LQG}}\right)^{-1}\mathbf{B}_{\mathrm{LQG}}^{\top}\mathbf{S}\mathbf{F}_{\mathrm{W}} \tag{67}$$

where the matrix $\mathbf{S}$ is given by

$$\mathbf{S} = \mathbf{F}_{\mathrm{LQG}}^{\top}\mathbf{S}\mathbf{F}_{\mathrm{LQG}} + \mathbf{T}_e - \mathbf{L}^{\top}\left(\mathbf{T}_u + \mathbf{B}_{\mathrm{LQG}}^{\top}\mathbf{S}\mathbf{B}_{\mathrm{LQG}}\right)\mathbf{L}, \tag{68}$$

where $\mathbf{S}[k]$ is the unique solution of the Ricatti equation. Similarly with [13, Eqs. 209 & 210], we define weighting matrices $\mathbf{T}_e$ and $\mathbf{T}_u$ as

$$\mathbf{T}_e = \mu_e \pi \operatorname{diag}\left(\beta, 1, \frac{1}{T}, \frac{2}{T^2}\right) \tag{69}$$

$$\mathbf{T}_u = \mu_u \pi \operatorname{diag}\left(\beta, 1, \frac{1}{T}, \frac{2}{T^2}\right). \tag{70}$$

3.1.3 Channel coefficients model

We assumed that the channel coefficients follow a random walk model, as it is also adopted in [8, Section 3] and [12, Section III]. Therefore, we have

$$\mathbf{x}_{\mathrm{H}}[k] = \mathbf{F}_{\mathrm{H}}\mathbf{x}_{\mathrm{H}}[k-1] + \mathbf{w}_{\mathrm{H}}[k-1] \tag{71}$$

where

$$\mathbf{x}_{\mathrm{H}}[k] = \left[\tilde{h}_0[k], \tilde{h}_1[k], \ldots, \tilde{h}_{\mathrm{L}}[k]\right]^{\top} \in \mathbb{R}^{L+1} \tag{72}$$

denotes the state variable, $\mathbf{F}_{\mathrm{H}} = \mathbf{I}_{L+1}$ is the state transition matrix, and

$$\mathbf{w}_{\mathrm{H}}[k] \sim \mathcal{CN}\left(\mathbf{0}_{L+1}, \mathbf{Q}_{H}\right) \tag{73}$$

represents the driving complex multivariate Gaussian noise of the channel coefficients random walk model, whose covariance is modeled as a diagonal matrix, such that $\mathbf{Q}_{\mathrm{H}} = \sigma_{\mathrm{H}}^2\mathbf{I}_{L+1}$, with $\sigma_{\mathrm{H}}^2 \in \mathbb{R}^{+}$. It is noteworthy that if we assumed the existence of cross-covariance terms in $\mathbf{Q}_{\mathrm{H}}$, it would be complex. However, as we assume that there is no cross-correlation between the individual driving noises, $\mathbf{Q}_{\mathrm{H}}$ is real.

---

[3] This solution can be readily given using the MATLAB's built-in DARE implementation [18]. For obtaining the matrix $\mathbf{L}$, use the following command line: `[~, L, ~] = idare(`$\mathbf{F}_{\mathrm{W}}$`, `$\mathbf{I}_4$`, `$\mathbf{T}_e$`, `$\mathbf{T}_u$`, `$\mathbf{0}_{4\times 4}$`, `$\mathbf{I}_4$`)`. You can verify that these inputs properly maps (68) and (67) to MATLAB'S Ricatti equation implementation.

3.2 Complex Extended Kalman Filter (CEKF) Algorithm for Signal Tracking and Channel Impulse Response Estimation

Unifying (65) and (71) in a single state variable and recalling (53), we can define the state and measurement equations that are used by the EKF as

$$
\begin{aligned}
\mathbf{x}[k] &= \mathbf{F}\mathbf{x}[k-1] + \mathbf{w}[k-1] \qquad (74)\\
\mathbf{z}[k] &= \underbrace{\sqrt{2P_s}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\,\mathbf{\Phi}_{pp}(\varepsilon_\tau[k],l)}_{\mathbf{h}(\mathbf{x}[k])} + \mathbf{v}_C[k]\\
&= \mathbf{h}(\mathbf{x}[k]) + \mathbf{v}_C[k]
\end{aligned}
$$

where

$$
\mathbf{x}[k] = [\delta\mathbf{x}_{\mathrm{W}}[k], \mathbf{x}_{\mathrm{H}}[k]]^\top \in \mathbb{C}^{L+5} \qquad (75)
$$

denotes the state variable,

$$
\mathbf{F} = \mathrm{blkdiag}(\mathbf{F}_{\mathrm{W}} - \mathbf{L}, \mathbf{F}_H) \in \mathbb{R}^{(L+5)\times(L+5)} \qquad (76)
$$

represents the state transition matrix, and

$$
\mathbf{w}[\mathbf{k}] = [\mathbf{w}_{\mathrm{W}}[k], \mathbf{w}_{\mathrm{H}}[k]] \sim \mathcal{CN}(\mathbf{0}_{L+5}, \mathbf{Q}) \qquad (77)
$$

characterizes the unified multivariate Gaussian noise, whose covariance matrix is given by

$$
\mathbf{Q} = \mathrm{blkdiag}(\mathbf{Q}_{\mathrm{W}}, \mathbf{Q}_{\mathrm{H}}) \in \mathbb{R}^{(L+5)\times(L+5)}. \qquad (78)
$$

Furthermore,

$$
\mathbf{h}(\mathbf{x}[k]) = \sqrt{2P_s}e^{j\varepsilon_\phi[kN]}\sum_{l=0}^{L}\tilde{h}_l[k]\,\mathbf{\Phi}_{pp}(\varepsilon_\tau[k],l) \qquad (79)
$$

is the measurements transition function. Additionally, for implementing the CEKF algorithm, it is necessary to calculate the jacobian of (79) evaluated for $\mathbf{x}[k] = \hat{\mathbf{x}}[k \mid k-1]$, which is given by

$$
\begin{aligned}
\tilde{\mathbf{H}}[k \mid k-1] &= \frac{\boldsymbol{\partial}\mathbf{h}(\mathbf{x}[k])}{\boldsymbol{\partial}\mathbf{x}[k]}\Big|_{\mathbf{x}[k]=\hat{\mathbf{x}}[k|k-1]}\\
&= [\mathbf{h}_\tau[k \mid k-1], \mathbf{h}_\phi[k \mid k-1], \mathbf{0}_{C\times 2}, \mathbf{H}_{\mathrm{H}}[k \mid k-1]]. \qquad (80)
\end{aligned}
$$

Here,

$$
\begin{aligned}
\mathbf{h}_\tau[k \mid k-1] &= \frac{\boldsymbol{\partial}\mathbf{h}(\hat{\mathbf{x}}[k \mid k-1])}{\partial\hat{\varepsilon}_\tau[k \mid k-1]}\\
&= \frac{\boldsymbol{\partial}}{\partial\hat{\varepsilon}_\tau[k \mid k-1]}\left\{\sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\sum_{l=0}^{L}\hat{\tilde{h}}_l[k \mid k-1]\,\mathbf{\Phi}_{pp}(\hat{\varepsilon}_\tau[k \mid k-1],l)\right\}\\
&= \sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\sum_{l=0}^{L}\hat{\tilde{h}}_l[k \mid k-1]\,\dot{\mathbf{\Phi}}_{pp}(\hat{\varepsilon}_\tau[k \mid k-1],l), \qquad (81)
\end{aligned}
$$

with

$$\dot{\mathbf{\Phi}}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], l\right) = \frac{\boldsymbol{\partial}\mathbf{\Phi}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], l\right)}{\partial\hat{\varepsilon}_\tau\left[k \mid k-1\right]} \tag{82}$$

being the partial derivative of the projected ahead measurements w.r.t. $\hat{\varepsilon}_\tau\left[k \mid k-1\right]$. We should then evaluate individually the derivatives of each element $\Phi_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right] + (l+m)\,T_s\right)$ of $\mathbf{\Phi}_{pp}\left(\varepsilon_\tau\left[k+1\right], l\right)$. Substituing $\Delta = \hat{\varepsilon}_\tau\left[k \mid k-1\right] + (l+m)\,T_s$ for τ in (6), we get

$$\Phi_{pp}\left(\Delta\right) = \begin{cases} 1 - \dfrac{|\Delta|}{T_c}, & |\Delta| \le T_c \\ 0, & \text{elsewhere} \end{cases}. \tag{83}$$

Deriving this expression w.r.t. to $\hat{\varepsilon}_\tau\left[k \mid k-1\right]$, applying the chain rule, and using the fact that $\frac{\partial\Delta}{\partial\hat{\varepsilon}_\tau[k|k-1]} = 1$, it yields

$$\begin{aligned} \frac{\partial\Phi_{pp}\left(\Delta\right)}{\partial\hat{\varepsilon}_\tau\left[k \mid k-1\right]} &= \frac{\partial\Phi_{pp}\left(\Delta\right)}{\partial\Delta}\frac{\partial\Delta}{\partial\hat{\varepsilon}_\tau\left[k \mid k-1\right]} = \frac{\partial\Phi_{pp}\left(\Delta\right)}{\partial\Delta} \\ &= -\frac{1}{T_c}\text{sign}\left(\Delta\right), 0 < |\Delta| < T_c = \begin{cases} -\dfrac{1}{T_c}, & 0 < |\Delta| < T_c \\ \dfrac{1}{T_c}, & -T_c < |\Delta| < 0 \\ 0, & \text{elsewhere} \end{cases} \end{aligned} \tag{84}$$

and non differentiable in $\Delta \in \{-T_c, 0, T_c\}$.

Moreover,

$$\begin{aligned} \mathbf{h}_\phi\left[k \mid k-1\right] &= \frac{\boldsymbol{\partial}\mathbf{h}\left(\hat{\mathbf{x}}\left[k \mid k-1\right]\right)}{\partial\hat{\varepsilon}_\phi\left[k \mid k-1\right]} \\ &= \frac{\boldsymbol{\partial}}{\partial\hat{\varepsilon}_\phi\left[k \mid k-1\right]}\left\{\sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\sum_{l=0}^{L}\hat{\hat{h}}_l\left[k \mid k-1\right]\mathbf{\Phi}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], l\right)\right\} \\ &= j\sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\sum_{l=0}^{L}\hat{\hat{h}}_l\left[k \mid k-1\right]\mathbf{\Phi}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], l\right) \end{aligned} \tag{85}$$

characterizes the partial derivative of the projected ahead measurement w.r.t. $\partial\hat{\varepsilon}_\phi\left[k \mid k-1\right]$.

Furthermore, we also have that

$$\frac{\boldsymbol{\partial}\mathbf{h}\left(\hat{\mathbf{x}}\left[k \mid k-1\right]\right)}{\partial\hat{\varepsilon}_\nu\left[k \mid k-1\right]} = \frac{\boldsymbol{\partial}\mathbf{h}\left(\hat{\mathbf{x}}\left[k \mid k-1\right]\right)}{\partial\hat{\varepsilon}_{\dot{\nu}}\left[k \mid k-1\right]} = \mathbf{0}_{C\times 1}, \tag{86}$$

and

$$\mathbf{H}_{\mathrm{H}}\left[k \mid k-1\right] = \left[\mathbf{h}_0\left[k \mid k-1\right], \mathbf{h}_1\left[k \mid k-1\right], \ldots, \mathbf{h}_r\left[k \mid k-1\right], \ldots, \mathbf{h}_L\left[k \mid k-1\right]\right], \tag{87}$$

where

$$\begin{aligned} \mathbf{h}_r\left[k \mid k-1\right] &= \frac{\boldsymbol{\partial}\mathbf{h}\left(\hat{\mathbf{x}}\left[k \mid k-1\right]\right)}{\partial\hat{\hat{h}}_r^*\left[k \mid k-1\right]} \\ &= \frac{\boldsymbol{\partial}}{\partial\hat{\hat{h}}_r^*\left[k \mid k-1\right]}\left\{\sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\sum_{l=0}^{L}\hat{\hat{h}}_l\left[k \mid k-1\right]\mathbf{\Phi}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], l\right)\right\} \\ &= \sqrt{2P_s}e^{j\hat{\varepsilon}_\phi[k|k-1]}\mathbf{\Phi}_{pp}\left(\hat{\varepsilon}_\tau\left[k \mid k-1\right], r\right). \end{aligned} \tag{88}$$

Therefore, we may apply the CEKF algorithm [9, Eqs. 14–18], using the same algorithmic structure shown in [17, Algorithm 1]

Algorithm 1: CEKF algorithm

Input: $\hat{\mathbf{x}}[1 \mid 0], \hat{\mathbf{P}}[1 \mid 0], \mathbf{z}[k], \mathbf{F}, \mathbf{Q}, \mathbf{R}$

Output: $\hat{\mathbf{x}}[k \mid -1] \;\; \forall\, k \in \{1, 2, \ldots, K\}$

`// Update step`

if $k > 1$ **then**

$\quad \tilde{\mathbf{H}}[k \mid k-1] \leftarrow \frac{\partial \mathbf{h}(\mathbf{x}[k])}{\partial \mathbf{x}[k]}\Big|_{\mathbf{x}[k]=\hat{\mathbf{x}}[k|k-1]}$

$\quad \mathbf{K}[k] \leftarrow \hat{\mathbf{P}}[k \mid k-1]\, \tilde{\mathbf{H}}^{\mathsf{H}}[k \mid k-1] \times$
$\quad\quad \times \left(\tilde{\mathbf{H}}[k \mid k-1]\, \hat{\mathbf{P}}[k \mid k-1]\, \tilde{\mathbf{H}}^{\mathsf{H}}[k \mid k-1] + \mathbf{R}\right)^{-1}$

$\quad \hat{\mathbf{x}}[k \mid k] \leftarrow \hat{\mathbf{x}}[k \mid k-1] + \mathbf{K}[k]\left(\mathbf{z}[k] - \mathbf{h}\left(\hat{\mathbf{x}}[k \mid k-1]\right)\right)$

$\quad \hat{\mathbf{P}}[k \mid k] \leftarrow \left(\mathbf{I}_{L+5} - \mathbf{K}[k]\, \tilde{\mathbf{H}}[k \mid k-1]\right) \hat{\mathbf{P}}[k \mid k-1]$

else

$\quad \hat{\mathbf{x}}[k \mid k] \leftarrow \hat{\mathbf{x}}[k \mid k-1]$

$\quad \hat{\mathbf{P}}[k \mid k] \leftarrow \hat{\mathbf{P}}[k \mid k-1]$

`// Project ahead`

$\hat{\mathbf{x}}[k+1 \mid k] \leftarrow \mathbf{F}\hat{\mathbf{x}}[k \mid k]$

$\hat{\mathbf{x}}[k+1 \mid k] \leftarrow \mathbf{F}\hat{\mathbf{x}}[k \mid k]\, \mathbf{F}^{\top} + \mathbf{Q}$

$k \leftarrow k + 1$

Figure 1 presents the CEKF algoritm in a high-level block diagram.

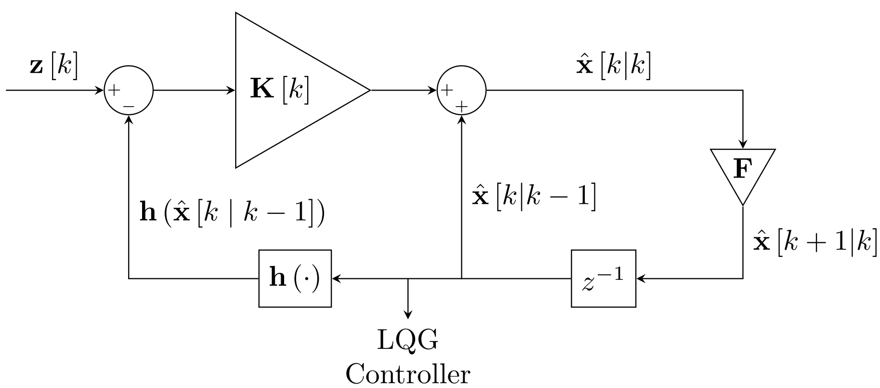

Figure 1: High-level block diagram representation of the CEKF implementation.

As commented beforehand, the Wiener model error-state provided by the CEKF algorithm is then used for feeding the LQG controller, that estimates the code-delay and carrier-phase evolution using the state-space model given by (62). Afterwards, a numerically controlled oscillator (NCO) receives $\hat{\mathbf{x}}_{\text{LQG}}[k]$ and outputs:

$$\hat{\mathbf{d}}^{*}[k] = \left[e^{-j\hat{\phi}[k]}, e^{-j\left(\hat{\phi}[k]+2\pi\hat{f}[k]T_s+\pi\hat{\dot{f}}[k]T_s^2\right)}, \ldots, e^{-j\left(\hat{\phi}[k]+2\pi\hat{f}[k](N-1)T_s+\pi\hat{\dot{f}}[k]((N-1)T_s)^2\right)}\right]^{\top}. \tag{89}$$

Additionally, the multicorrelator (39) selects $\hat{\tau}_{\text{T}}[k]$ from $\hat{\mathbf{x}}_{\text{LQG}}[k]$ and generates the equidistantly delayed PR-code replicas for performing correlation with the received signal in the next time step.

Figure 2 shows the full proposed system for code-delay, carrier-phase and channel estimation.

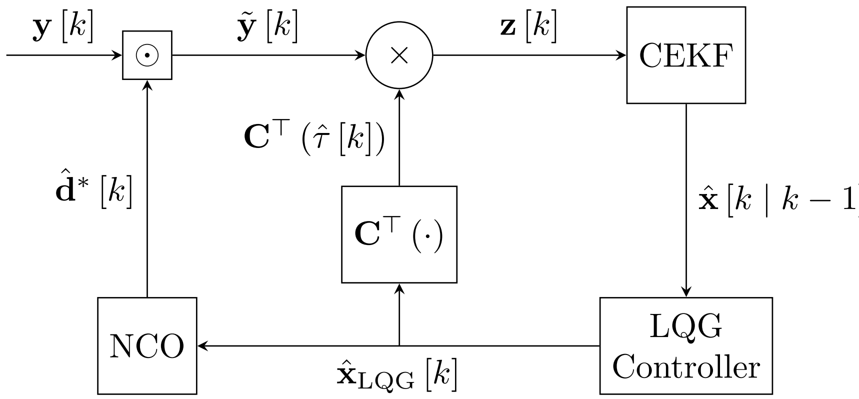

Figure 2: High-level block diagram representation of the full system implementation.

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